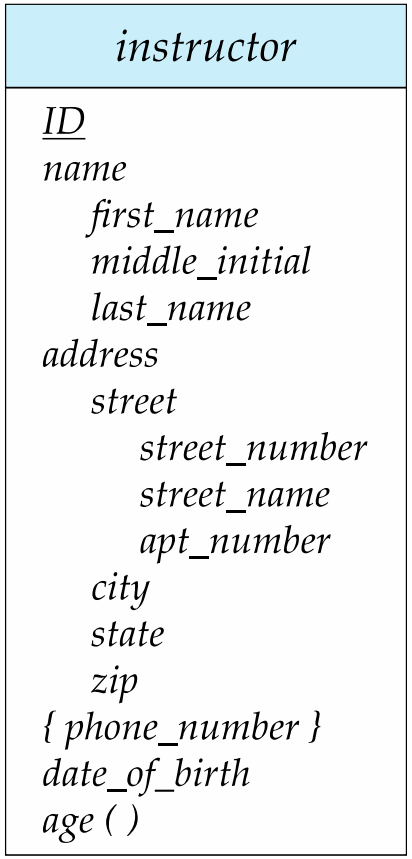
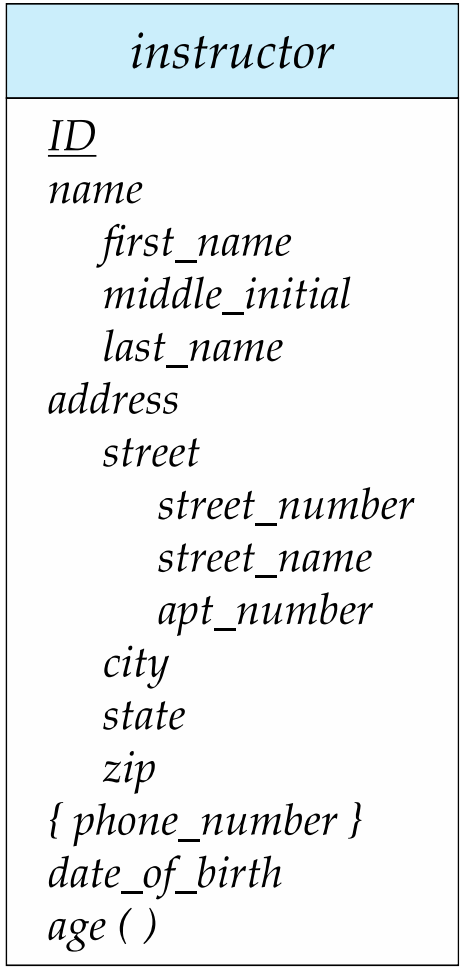
ER to Schema

Representing Complex Attributes in ER Diagram





Representation of Composite Attributes

- Composite attributes are flattened out by creating a separate attribute for each component attribute
- Ignoring multivalued attributes, the extended instructor schema is
  - *instructor(ID,*
    - *first_name,*
    - *middle_initial,*
    - *last_name,*
    - *street_number,*
    - *street_name,*
    - *apt_number,*
    - *city,*
    - *state,*
    - *zip_code,*
    - *date_of_birth)*

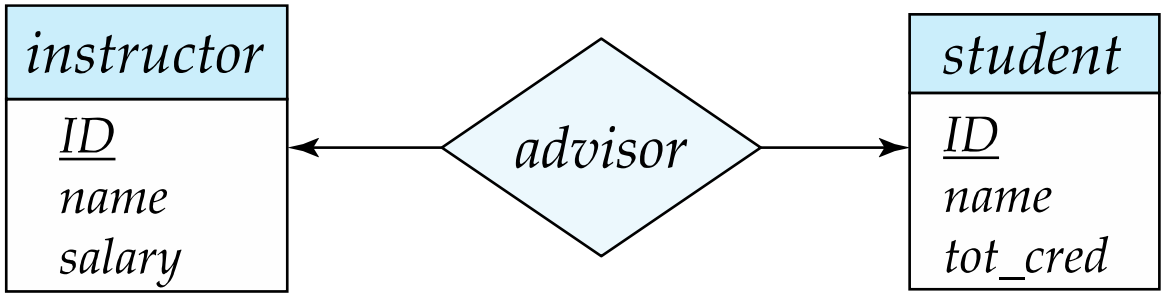
instructor (ID, first_name, middle_initial, last_name, street_number, street_name, apt_number, city, state, postal_code, date_of_birth)

instructor_phone (ID, phone_number)

Cardinality Constraints in ER Diagram

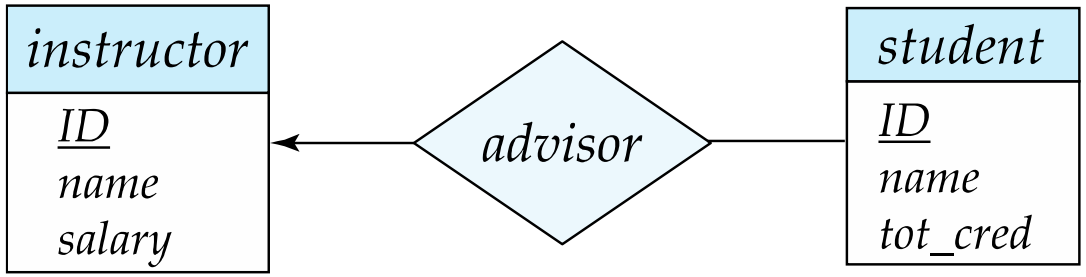
- We express cardinality constraints by drawing either a directed line (→), signifying "one," or an undirected line (—), signifying "many," between the relationship set and the entity set.

- One-to-one relationship between an *instructor* and a *student* :
  - A student is associated with at most one *instructor* via the relationship *advisor*
  - A *student* is associated with at most one *department* via *stud_dept*



One-to-Many Relationship

- One-to-many relationship between an *instructor* and a *student*
  - An instructor is associated with several (including 0) students via *advisor*
  - A student is associated with at most one instructor via advisor,



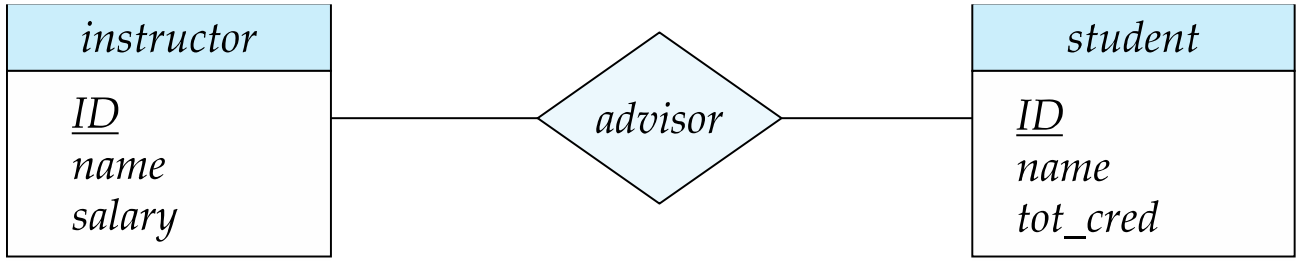
Many-to-One Relationships

- Many-to-one relationship between an *instructor* and a *student*
  - An instructor is associated with at most one student via *advisor*,
  - A student is associated with several (including 0) instructors via *advisor*



Many-to-Many Relationship

- An instructor is associated with several (possibly 0) students via *advisor*
- A student is associated with several (possibly 0) instructors via *advisor*



Total and Partial Participation

- **Total participation** (indicated by double line): every entity in the entity set participates in at least one relationship in the relationship set

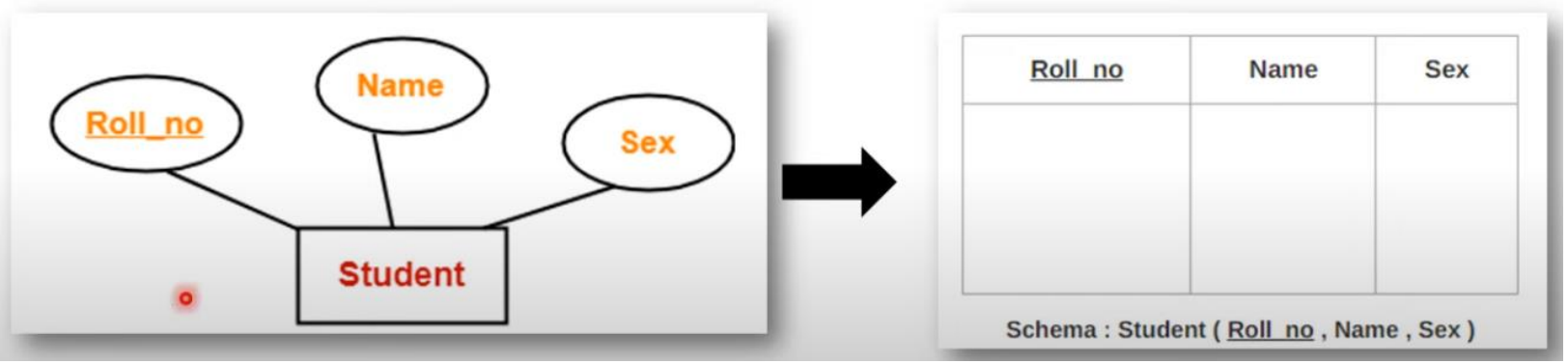


participation of *student* in *advisor* relation is total

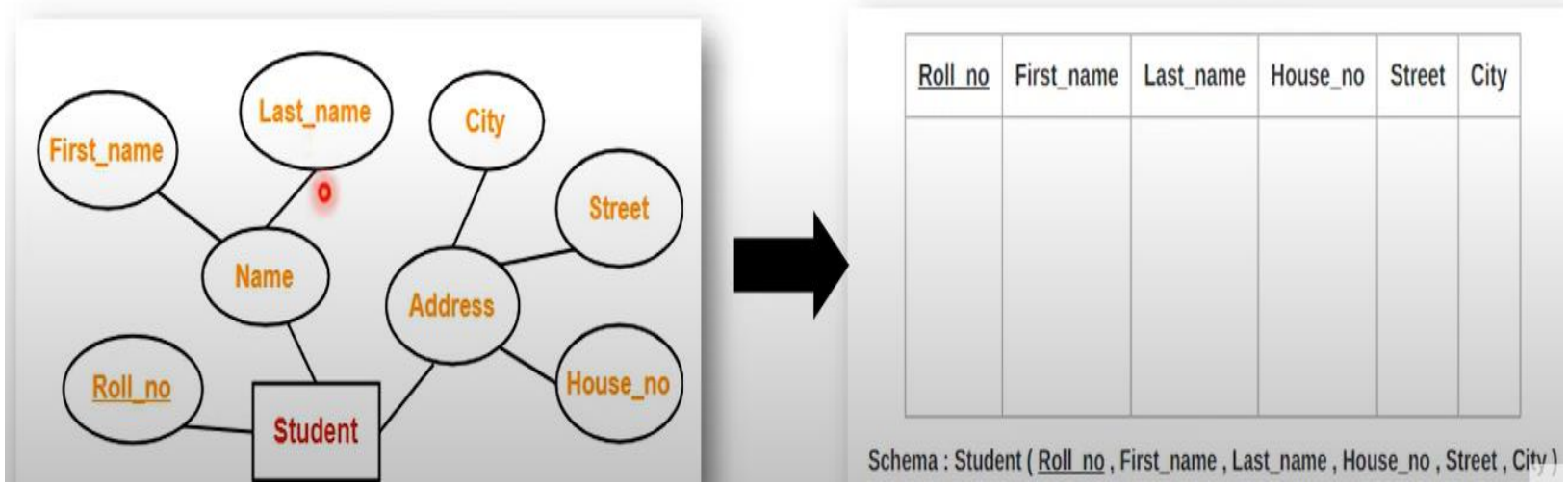
  - Every *student* must have an associated instructor

- **Partial participation**: some entities may not participate in any relationship in the relationship set
  - Example: participation of *instructor* in *advisor* is partial

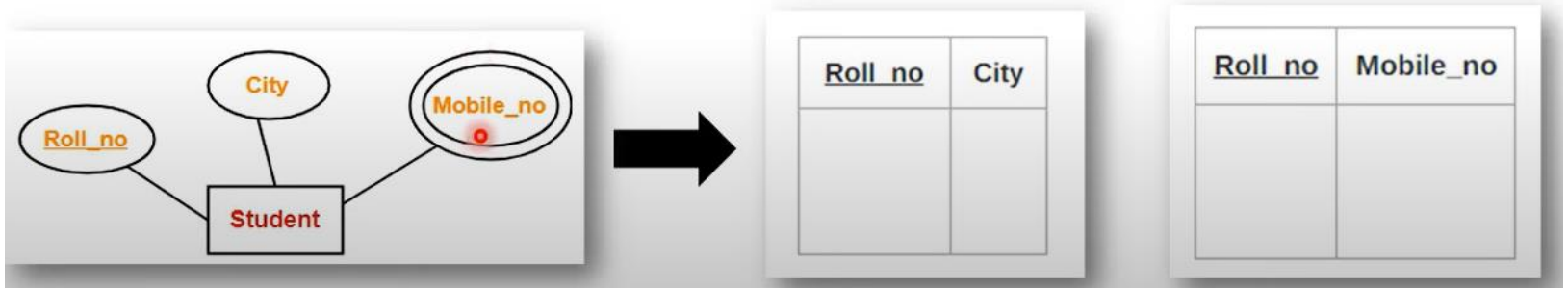
Strong Entity Set with Simple Attributes



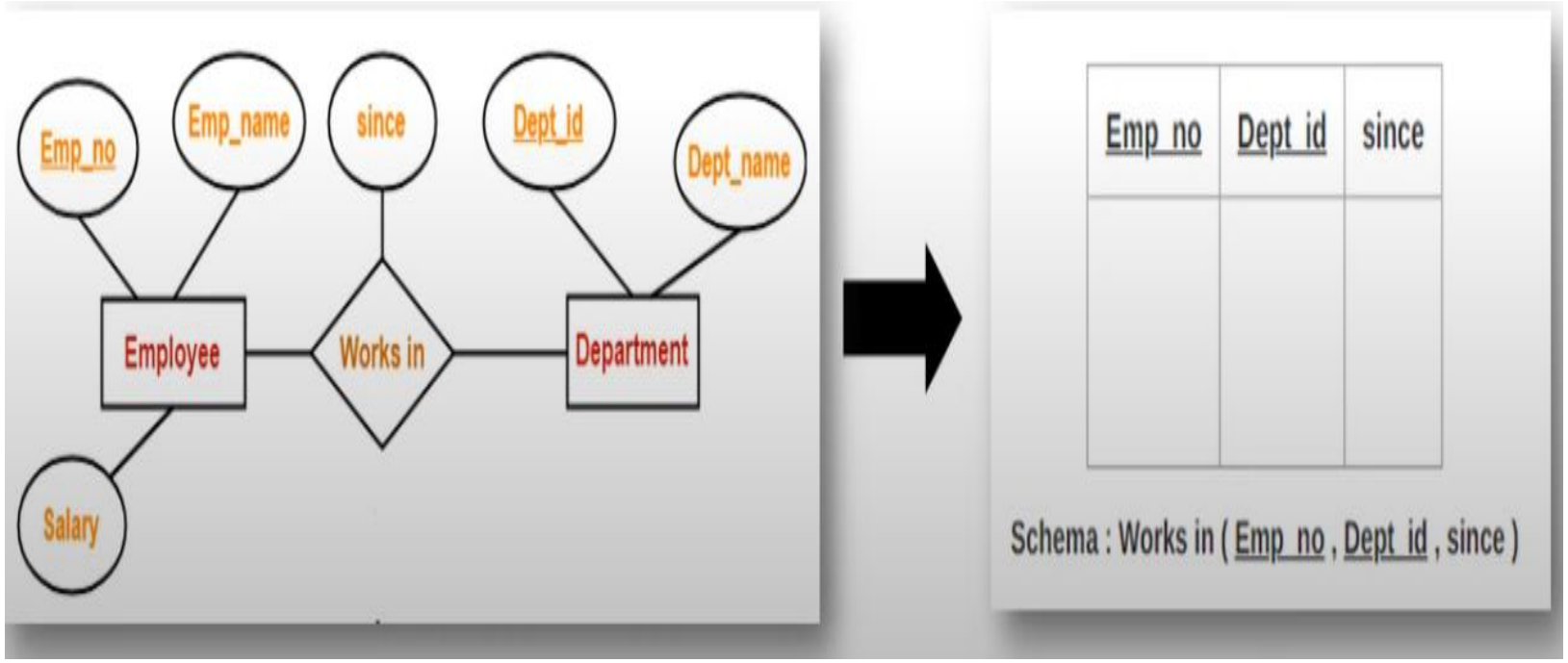
Strong Entity Set with Composite Attributes



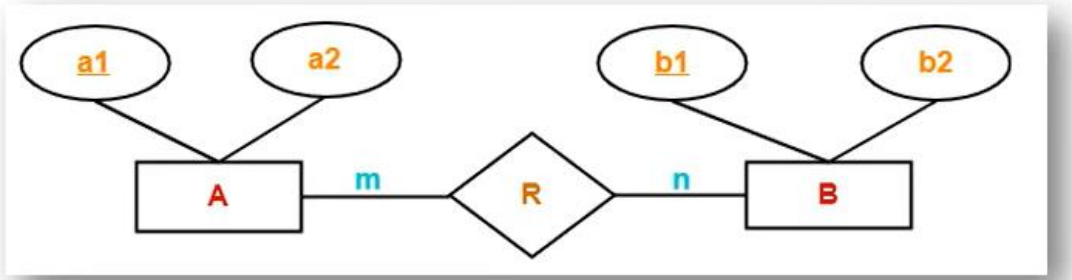
Strong Entity Set with Multivalued Attributes



Relationship to Relation



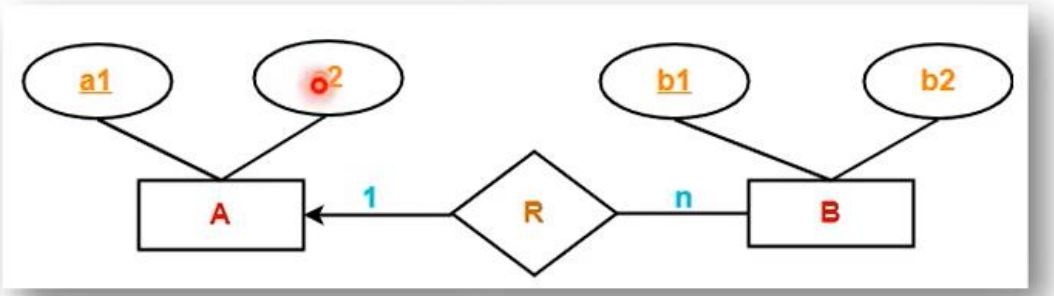
✓**Case-1:** For Binary Relationship With Cardinality Ratio **m:n**



Here, three tables will be required-

1. A (a1 , a2)
2. R (a1 , b1)
3. B (b1 , b2)

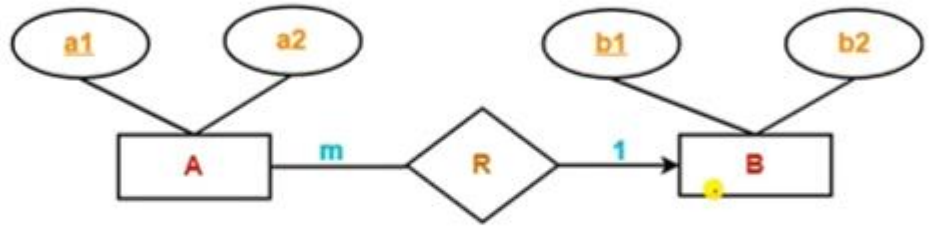
✓**Case-2:** For Binary Relationship With Cardinality Ratio **1:n**



Here, two tables will be required-

1. A (a1 , a2)
2. BR (a1 , b1 , b2)

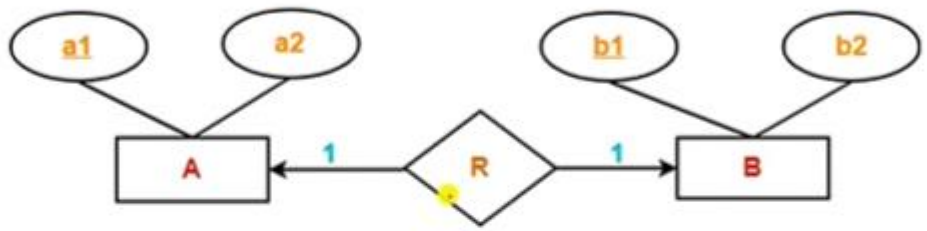
Case-3: For Binary Relationship with Cardinality Ratio m:1



In ***Many-to-One relationship***, **two tables** will be required-

1. **AR (a1 , a2 , b1)**
2. **B (b1 , b2)**

Case-4: For Binary Relationship with Cardinality Ratio 1:1



In ***One-to-One relationship***, **two tables** will be required. Either combine 'R' with 'A' or 'B'

Way-01:

1. **AR (a1 , a2 , b1)**
2. **B (b1 , b2)**

Way-02:

1. **A (a1 , a2)**
2. **BR (b1 , b2, a1)**

Thumb Rules to Remember:

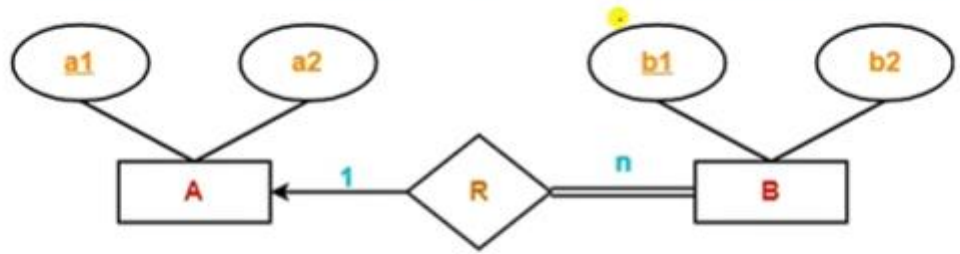
(For determining minimum number of tables)

- **While determining the minimum number of tables required for binary relationships with given cardinality ratios**, following thumb rules must be kept in mind-
  - For binary relationship with cardinality ration **m : n** ,
    - *Separate and individual tables* will be drawn for each entity set and relationship (i.e. **three tables** will be required).
  - For binary relationship with cardinality ratio either **m : 1 or 1 : n** ,
    - Always remember "many side will consume the relationship" i.e. a combined table will be drawn for many side entity set and relationship set (i.e. **Two tables** will be required).
  - For binary relationship with cardinality ratio **1 : 1** ,
    - **Two tables** will be required. You can combine the relationship set with any one of the entity sets.

Rule-6: For Binary Relationship with Both Cardinality Constraints and Participation Constraints

- Cardinality constraints will be implemented as discussed in Rule-5.
- Because of the *total participation constraint*, foreign key acquires **NOT NULL** constraint i.e. now foreign key can not be null.
- Two Cases:
  - **Case-1**: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side
  - **Case-2**: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

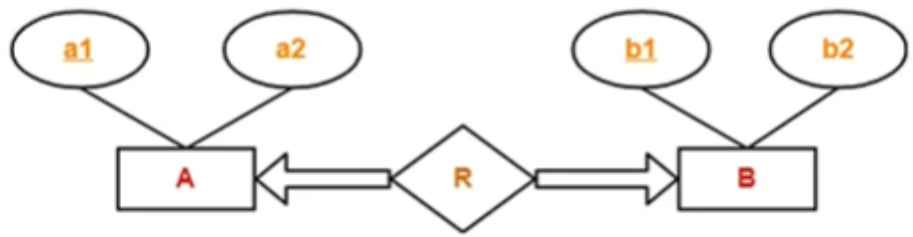
Case-1: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side



- Because cardinality ratio = 1 : n , we will combine the entity set B and relationship set R.
- Then, **two tables** will be required-
  1. **A (a1 , a2)**
  2. **BR (b1 , b2 , a1)**

Case-2: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

- If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using **only single table.**



- Here, Only **one table** is required.
  1. **ARB (a1 , a2 , b1 , b2)**

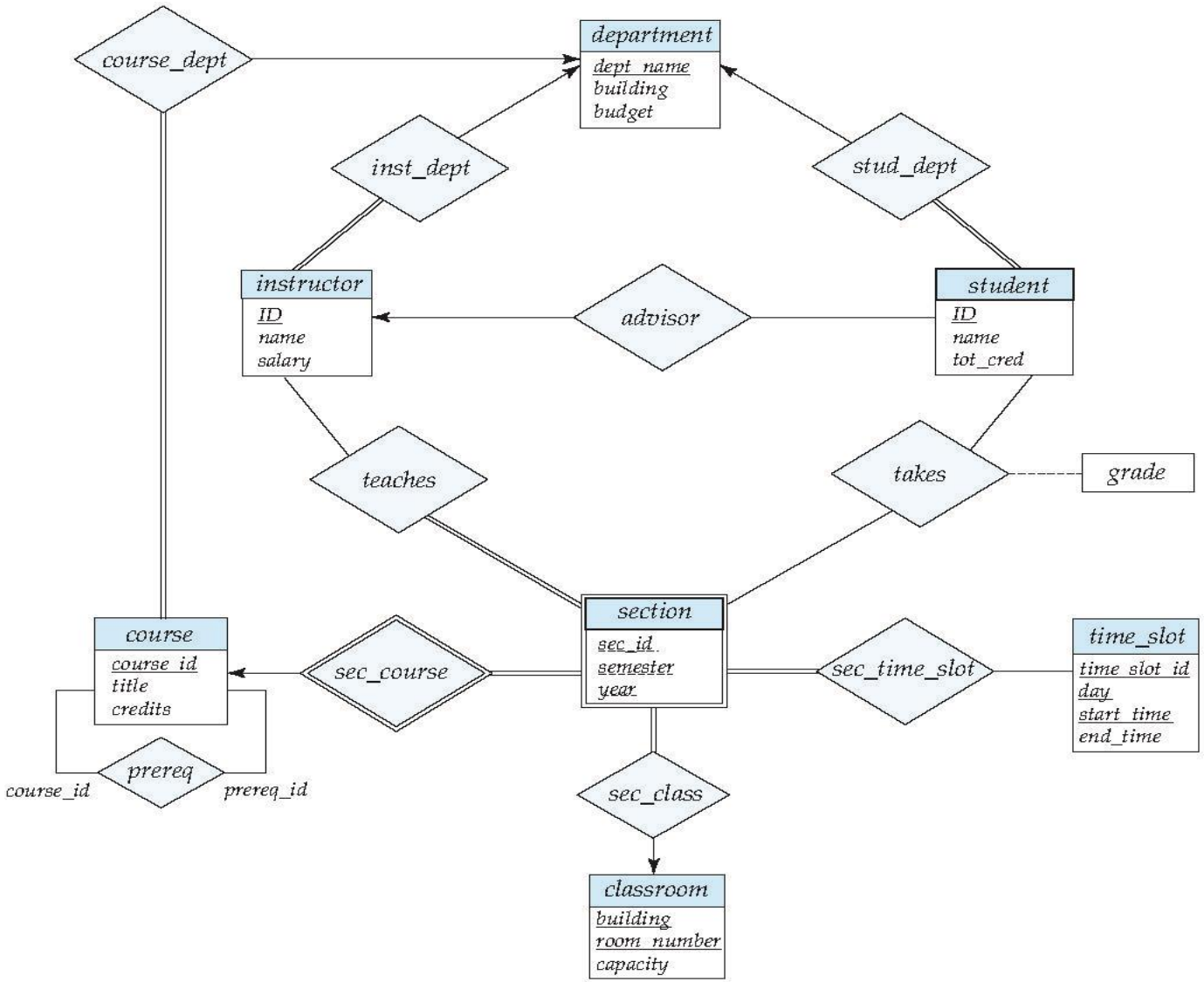
Rule-7: For Binary Relationship with Weak Entity Set

- Weak entity set always appears in association with identifying relationship with **total** participation constraint and there is always **1: n** relationship from identifying entity set to weak entity set.



- Here, **two tables** will be required-
  1. **A (a1 , a2)**
  2. **BR (a1 , b1 , b2)**

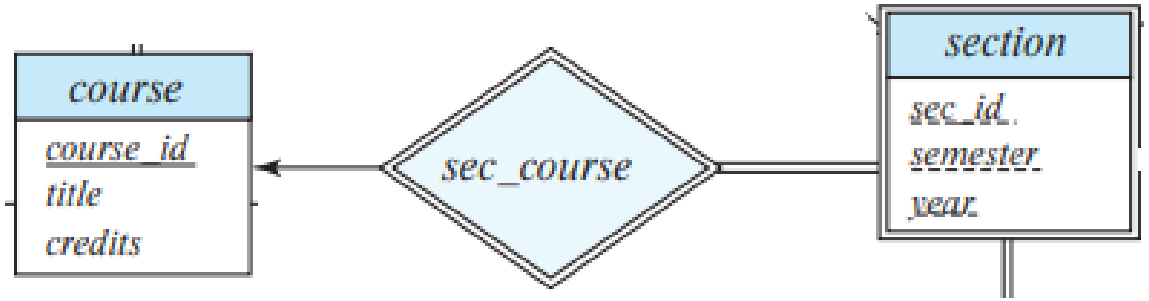
E-R Diagram for a University Enterprise



Strong and weak entity schema

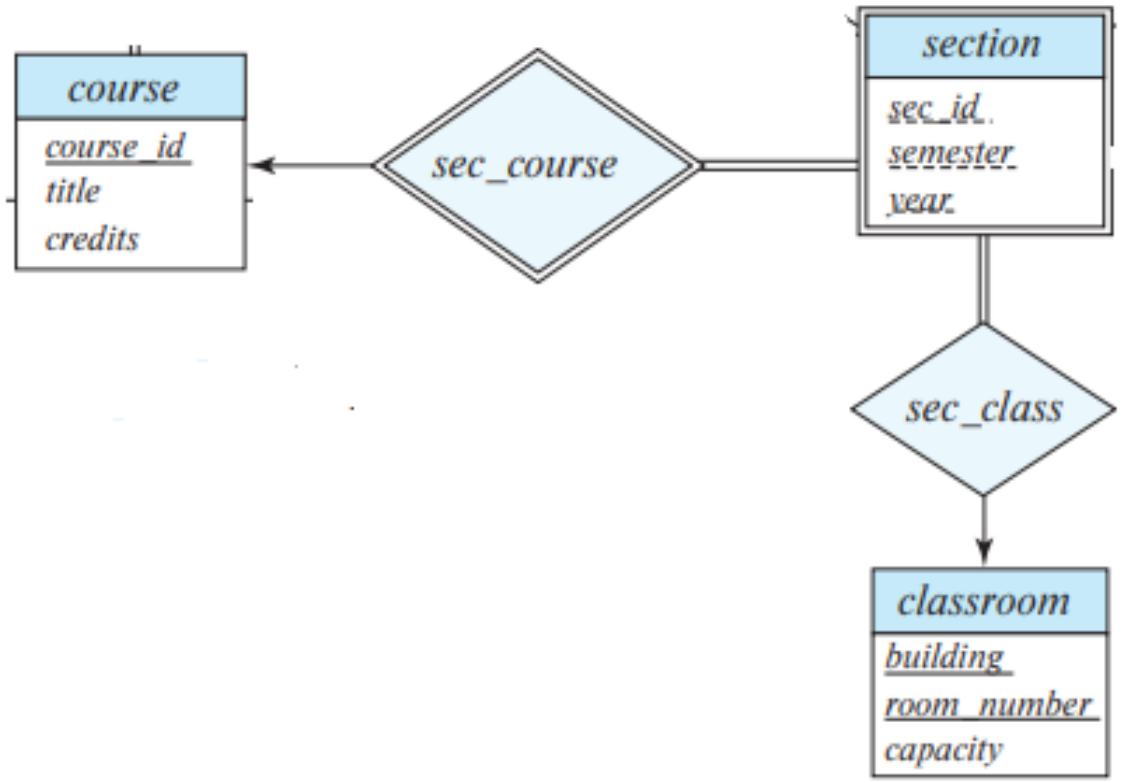
classroom(*building*, *room_number*, *capacity*)
department(*dept_name*, *building*, *budget*)
course(*course_id*, *title*, *credits*)
instructor(*ID*, *name*, *salary*)
student(*ID*, *name*, *tot_cred*)

Figure 6.16 Schemas derived from the entity sets in the E-R diagram in Figure 6.15.



section (*course_id*, *sec_id*, *semester*, *year*)

Relationship schema



sec class. The schemas section and classroom correspond to the entity sets A and B, respectively. Thus, the schema sec class can be combined with the section schema. The resulting section schema consists of the attributes **{course id, sec id, semester, year, building, room number}**